

Palladium-Catalyzed Cross-Coupling Reaction of Functionalized Aryl- and Heteroaryl-bismuthanes with 2-Halo(or 2-Triflyl)azines and -diazines

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The palladium-catalyzed cross-coupling of highly functionalized organobismuthanes with 2-halo(or 2-triflyl)pyridines, -pyrimidines, -pyrazines, and -pyridazines is reported. The reaction tolerates numerous functional groups, including al-

dehydes. The synthesis of a shelf-stable (formylphenyl)bismuth reagent and its use in a cross-coupling reaction is also described.

Introduction

Azines and diazines are ubiquitous in pharmaceutical industry. Compounds that incorporate these heterocycles have been found to possess antimicrobial, antiviral, antioxidant, antitumoral, anti-inflammatory, and other activities.^[1] Azines and diazines have also been used in the preparation of materials that show interesting electrical and optical properties.^[2] Consequently, access to efficient and general methods that allow their preparation is of utmost importance.^[3] To be applicable to the context of drug discovery, these methods should tolerate the presence of functional groups that can serve as handles for further transformation of the molecule.^[4] Ideally, the protocol should also involve reagents that can be easily manipulated and should operate under simple conditions that can be amenable to parallel chemistry.^[5]

The metal-catalyzed cross-coupling reaction between halogenated electrophiles and organometallic reagents has evolved into an extremely powerful approach for the installation of aromatic and heteroaromatic units onto a variety of scaffolds.^[6] The cross-coupling of aryl- and heteroaryl-metal compounds with 2-halo(or 2-triflyl)azines and -diazines constitutes an attractive approach to access the corresponding 2-arylheterocycles. A few examples of cross-coupling reactions between organometallic reagents and 2-haloazines and -diazines have been reported.^[7,8] However, many of these methods show modest substrate scope and require air- and moisture-sensitive reagents or complex catalysts. More importantly, limited functional group tolerance is often observed, particularly in the case of reactive

organometallic species such as organomagnesium reagents.^[9] Although organostannanes are more tolerant to acidic and electrophilic groups,^[10] their high toxicity combined with the fact that only one aryl group is usually transferred make them considerably less attractive. Coupling of 2-haloazines and -diazines with organozinc reagents offers a great solution to issues of functional group tolerance.^[11] However, these species must imperatively be manipulated under inert and anhydrous conditions, which complicates their use in parallel chemistry. To the best of our knowledge, the most general methods for the transfer of aryl and heteroaryl units onto azines and diazines are those of Knochenel^[12,13] and Quéguiner.^[14] However, as these procedures involve organomagnesium reagents, sensitive groups such as aldehydes and ketones cannot be present on either coupling partner.

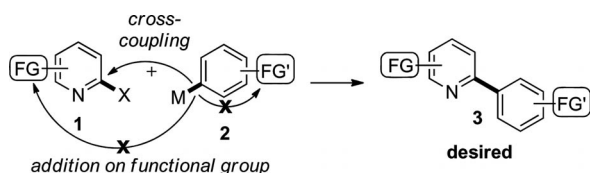
Organobismuthanes have found increasing use in bond formation owing to their unique properties and reactivity.^[15] These organometallic reagents can be easily prepared,^[16] chromatographed, and stored in air at room temperature. In addition, their low toxicity makes them extremely attractive species for methodology development. The low polarity of the C–Bi bond provides a moderate reactivity for this class of reagents, which thus allows the presence of numerous reactive or acidic functional groups on the electrophilic partner. Cross-coupling reactions involving triphenylbismuth was first reported by Barton et al. in 1988.^[17] Recently, the groups of Rao,^[18] Tanaka,^[19] and others^[20] have extended the scope of this reaction to include other organobismuthanes. However, to the best of our knowledge, the full potential of organobismuthanes in the transfer of highly functionalized groups onto 2-halo(or 2-triflyl)azines and -diazines has never been demonstrated.

We reported over the past years applications of trivalent^[21,22] and pentavalent organobismuthanes^[23] for the formation of C–C and C–N bonds. Recently, we disclosed the first palladium-mediated cross-coupling reaction of tri-

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alkylbismuth reagents with a variety of scaffolds possessing reactive functional groups, including one example involving a 2-chloropyridine.^[24] We report herein our study on the cross-coupling reaction of 2-halo(or 2-triflyl)pyridines, -pyrazines, -pyrimidines, and -pyridazines with functionalized aryl- and heteroaryl-bismuthanes.

To have high value, we aimed for a protocol that would involve a simple catalyst, shelf-stable organometallic species and that would permit the cross-coupling between two units that both possess sensitive functional groups. Functionalized organometallics are extremely valuable species in organic synthesis.^[25] However, one of the major issue consists in designing an organometallic reagent that will selectively react at the desired position of the heterocycle without attacking the functional groups present on the electrophile or on the organometallic species (Scheme 1). We report herein our results on the preparation of organobismuthanes that bear functionalized groups, that are shelf-stable and that can be used in palladium-catalyzed C–C bond-forming reactions.



Scheme 1. Chemoselectivity of a cross-coupling reaction between functionalized organometallic species and 2-halopyridines. FG, FG' = functional group; M = metal; X = halide or pseudohalide.

Results and Discussion

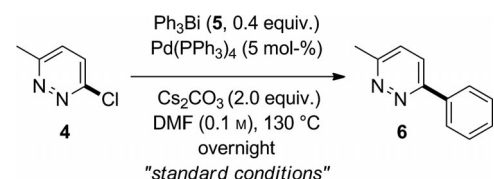
Before exploring functional group compatibility, we began by optimizing the reaction conditions for the coupling of triphenylbismuth (0.4 equiv.) with chlorodiazine **4**. Using conditions that we previously reported for the coupling of

tricyclopropylbismuth^[21] and other trialkylbismuthanes,^[24] we obtained 65% yield of desired product **6** (Table 1, Entry 1).

To assure that all three aryl groups were effectively transferred, we verified that the yield of product **6** was the same upon using 0.7 and 1.1 equiv. of triphenylbismuth (Table 1, Entries 2 and 3). The fact that similar yields were obtained demonstrate that all three phenyl groups were indeed transferred during the process. This well-established characteristic of triaryl-bismuth reagents makes them more atom-economical than other reagents such as aryltrialkyltin species. Performing the reaction at higher temperatures under thermal (Table 1, Entry 4) or microwave conditions (Table 1, Entry 5) did not lead to any improvement in the yield of the reaction. Doubling the catalyst loading (Table 1, Entry 6) or changing the system to favor oxidative addition in C–Cl bonds (Table 1, Entry 7; Cy = cyclohexyl) gave a performance similar to that of the standard conditions. Surprisingly, attempts at using Buchwald phosphanes did not provide any improvement in the yield of the reaction.^[26] Although changing the solvent to *N*-methylpyrrolidone (NMP) or toluene gave a lower yield of the coupling product (Table 1, Entries 8 and 9), we found that the use of a 4:1 mixture of DMF/hexamethylphosphoramide (HMPA) gave a noticeable improvement in the efficiency of the reaction (Table 1, Entry 10). To evaluate the importance of the base, we then conducted a reaction without cesium carbonate and obtained the desired product in 42% yield (Table 1, Entry 11); this suggests that – although the base is not essential – it is still required to achieve optimal yields. The results of the optimization of the reaction conditions demonstrate that the cross-coupling transformation can be accomplished by using very simple conditions and requires only approximately one third of an equivalent of the triaryl-bismuth reagent.

We next studied the substrate scope by using pyrazines **7**, pyrimidines **8**, pyridines **9**, and pyridazines **10** (Table 2).

Table 1. Optimization of the reaction conditions.

		
"standard conditions"		
Entry	Change from "standard conditions"	Isolated yield [%] ^[a]
1	no change	65
2	0.7 equiv. Ph ₃ Bi instead of 0.4 equiv.	62
3	1.1 equiv. Ph ₃ Bi instead of 0.4 equiv.	60
4	150 °C instead of 130 °C	50
5	170 °C (microwave) instead of 130 °C	36
6	10 mol-% Pd(PPh ₃) ₄ instead of 5 mol-%	60
7	Pd(OAc) ₂ (5 mol-%)/PCy ₃ (10 mol-%) instead of Pd(PPh ₃) ₄	50
8	NMP instead of DMF	10
9	toluene instead of DMF	50
10	DMF/HMPA (4:1) instead of DMF	72
11	no base	42

[a] Yield of isolated pure product **6**.

To compare the relative reactivity of these four classes of electrophiles, we performed all cross-coupling reactions by using triphenylbismuth under the optimal conditions found

in Table 1 (Entry 1, 2, or 10). We found that pyrazines smoothly underwent the cross-coupling reaction with Ph_3Bi to provide the desired products in good to excellent yields

Table 2. Coupling of triphenylbismuth with halo(or triflyl)azines and -diazines.

$\text{X} = \text{Cl, Br, OTf}$

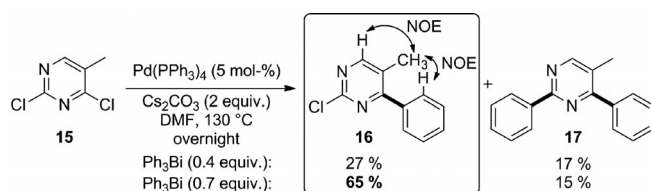
Entry	Conditions ^[a]	Electrophile	Product	Yield [%] ^[b]	Entry	Conditions ^[a]	Electrophile	Product	Yield [%] ^[b]
1	A			90	11	A			42
2	A			81	12	B			80
3	A			62	13	B			85
4	B			72	14	B			66
5	A ^[c]			62	15	B			80
6	A			65	16	A			45
7	B			90	17	A			62
8	A			50	18	A			30
9	C			76	19	A			90
10	A			50					

[a] Conditions A: Ph_3Bi (0.4 equiv.), $\text{Pd}(\text{PPh}_3)_4$ (5 mol-%), Cs_2CO_3 (2 equiv.), DMF, 130 °C, overnight. Conditions B: Ph_3Bi (0.7 equiv.), $\text{Pd}(\text{PPh}_3)_4$ (5 mol-%), Cs_2CO_3 (2 equiv.), DMF, 130 °C, overnight. Conditions C: Ph_3Bi (0.4 equiv.), $\text{Pd}(\text{PPh}_3)_4$ (5 mol-%), Cs_2CO_3 (2 equiv.), DMF/HPMA (4:1), 130 °C, overnight. [b] Yield of isolated pure product. [c] 10 mol-% of $\text{Pd}(\text{PPh}_3)_4$ instead of 5 mol-%.

(Table 2, Entries 1–4). Electron-withdrawing and electron-donating groups had little impact on the outcome of the reaction (Table 2, Entry 3 vs. 4). We next turned our attention to pyrimidines and obtained a good yield of the desired product upon coupling triphenylbismuth with 2-chloro-5-ethylpyrimidine (**8a**; Table 2, Entry 5). We demonstrated previously that 2-chloropyridines are very competent electrophiles in cross-coupling reactions with trialkylbismuth reagents.^[24] We wanted to extend the scope of the reaction to include bromo and triflyl derivatives. Our results show that 2-bromopyridines are generally more reactive than the 2-chloro analogues (e.g., **9b** vs. **9a**, **9e** vs. **9f**), which is consistent with a palladium-mediated pathway that involves oxidative addition in the C–X bond. In fact, 2-fluoropyridines failed to react under our conditions. Gratifyingly, 2-triflyl derivatives were found to have similar reactivity to the chloro analogues (e.g., **9i** vs. **9a**). These substrates can be easily accessed from the corresponding 2-hydroxypyridines. Finally, we performed the coupling reaction on 2-chloropyridazine (**10a**) and obtained desired product **14a** in excellent yield (Table 2, Entry 19).

These results show that a great diversity of functional groups are tolerated in this reaction, including amines (Table 2, Entry 4), esters (Table 2, Entries 8, 10, and 11), aldehydes (Table 2, Entry 12), nitro groups (Table 2, Entry 16), nitriles (Table 2, Entries 3 and 17), and trifluoromethyl groups (Table 2, Entries 13 and 15). Even acidic and enolizable groups such as methyl ketones (Table 2, Entry 9) and alcohols (Table 2, Entry 18) are tolerated in this reaction. This is in sharp contrast to coupling reactions involving organomagnesium reagents in which these types of groups are generally incompatible with the very reactive nature of these organometallic species.

We next studied the regioselectivity of the cross-coupling reaction involving 2,4-dichloro-5-methylpyrimidine (**15**, Scheme 2). Upon using 0.4 equiv. of triphenylbismuth, a mixture of mono- and diphenyl products **16** and **17** was obtained in modest yield, and there was a slight preference for the compound resulting from the coupling at the 4-position (as indicated by NOE experiments). Fortunately, the yield of monophenyl compound **16** could be considerably ameliorated, without increasing the amount of diphenyl product **17**, by using 0.7 equiv. of triphenylbismuth.



Scheme 2. Regioselectivity of cross-coupling between triphenylbismuth and 2,4-dichloro-5-methylpyrimidine (**15**).

To explore the full potential of organobismuthanes in the transfer of highly functionalized groups, we next prepared functionalized organobismuth reagents **18a–j** and coupled

them with 2-chloroquinoxaline (**7a**, Table 3). In general, similar yields were obtained for triaryl-bismuthanes possessing electron-neutral (Table 3, Entries 1 and 7), electron-withdrawing (Table 3, Entries 3, 5, and 6), and electron-donating groups (Table 3, Entries 4 and 8). Under our standard conditions, *para*- (Table 3, Entries 1, 3, 6, and 8) and *meta*- (Table 3, Entries 4, 5, and 7) -substituted aryl groups were efficiently transferred by using the corresponding organobismuthanes. To our surprise, the coupling of *ortho*-substituted aryl groups proved to be more difficult than anticipated, as only a modest yield of the desired product was obtained (Table 3, Entry 2). Acid-sensitive groups such as acetals were well tolerated in this protocol (Table 3, Entries 7 and 8). Reactive electrophilic groups such as nitriles (Table 3, Entry 5) and esters (Table 3, Entry 6) were unaffected during the transformation, which demonstrates the mildness of the cross-coupling reaction with organobismuthanes. Finally, a 2-thienyl (Table 3, Entry 9) and a 3-quinolinyl (Table 3, Entry 10) unit were installed uneventfully on the quinoxaline by using our procedure.

Having established general conditions for the coupling of functionalized organobismuthanes with 2-haloazines and -diazines, we wanted to extend our methodology to the transfer of an aryl fragment that would bear a more reactive functional group such as an aldehyde. This particular functional group is highly reactive and is therefore incompatible with strong organometallic species such as organomagnesium reagents. However, owing to the mild reactivity of the C–Bi bond, we postulated that the aldehyde would be tolerated during the formation and the cross-coupling reaction of the organobismuthane. Only two examples of (formylaryl)bismuthanes have been reported in the literature, but the yields for their synthesis were low, and the species were not used in subsequent coupling reactions.^[27]

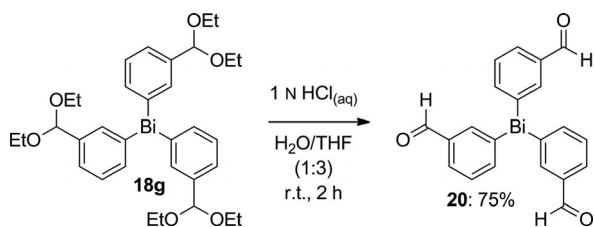
To test the compatibility of aldehydes with organobismuthanes in the context of a palladium cross-coupling reaction, we sought to prepare a formylphenylbismuth reagent directly from **18g**. The question we asked ourselves was how to liberate the aldehyde without breaking the C–Bi bond. To our great pleasure, tris(3-formylphenyl)bismuthane (**20**) was obtained in 75% yield simply by exposing **18g** to aqueous acidic conditions (Scheme 3). These results show that the C–Bi is strong enough to resist protic conditions. This reaction was performed successfully on a gram scale, and no decomposition was noticed after 1 month upon storing the reagent at room temperature in air. A limited number of organometallic reagents bearing formylaryl groups have been described in the literature,^[28,29] and to the best of our knowledge, only three reports of the hydrolysis of acetals directly on the organometallic reagent could be found, those being with germanium,^[30] antimony,^[31] and mercury.^[32]

To unequivocally confirm the structure of this new organometallic reagent, compound **20** was recrystallized. X-ray diffraction analysis revealed that the compound is pyramidal and has C–Bi–C angles of 92.02° and C–Bi bond lengths of 2.266 Å (Figure 1). The structure clearly shows the presence of three aryl groups each bearing a formyl moiety.

Table 3. Cross-coupling of functionalized triaryl- and triheteroarylbi-muthanes **18a–j** with 2-chloroquinoline (**7a**).

Entry	Organobismuthane	Product	Yield [%] ^[a]	Entry	Organobismuthane	Product	Yield [%] ^[a]
1			86	7			98
2			20	8			80
3			90	9			85
4			97	10			45
5			84				
6 ^[b]			80				

[a] Yield of isolated pure product. [b] 10 mol-% of Pd(PPh₃)₄ and 0.7 equiv. of Ar₃Bi were used.

Scheme 3. Synthesis of tris(3-formylphenyl)bismuthane (**20**).

We next verified the transferability of the formylphenyl group of **20** by conducting the cross-coupling reaction with pyrazine **7a** under our standard conditions and obtained the corresponding cross-coupling product **21** in 80% isolated yield (Scheme 4).

Isolated examples of cross-coupling reactions leading to the transfer of a formylaryl group on azines and diazines have been reported.^[33–36] However, to the best of our knowledge, the only general methods for the coupling of

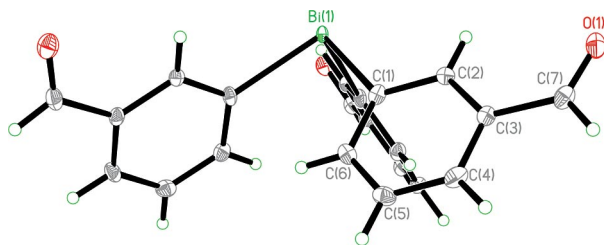
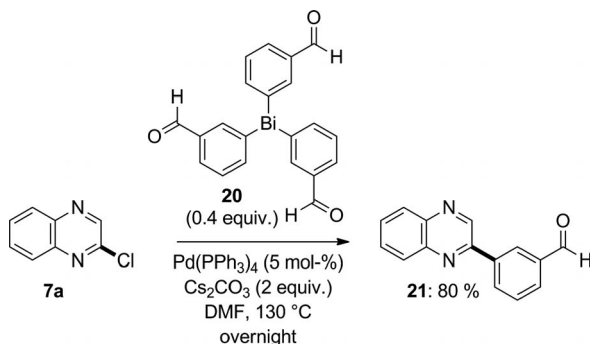


Figure 1. ORTEP diagram of compound **20**. Thermal ellipsoids are shown at the 50% probability level.



Scheme 4. Cross-coupling reaction between (formylphenyl)bismuth **20** and 2-chloroquinoline (**7a**).

highly functionalized aryl groups (including formylaryl) are those developed by Knochel that involve organozinc^[37] and organoindium reagents.^[38] Contrary to organozinc species, organobismuthanes are not moisture-sensitive and can be manipulated in air.

Conclusions

We have developed an efficient protocol for the cross-coupling of functionalized aryl- and heteroaryl-bismuth reagents with 2-halo(or 2-triflyl)pyridines, -pyrimidines, -pyrazines, and -pyridazines bearing reactive moieties. This method involves shelf-stable organobismuthanes and proceeds under simple conditions to provide the desired coupling products in good to excellent yields. All three aryl groups are transferred from the bismuthane in the course of the coupling process. The coupling on 2,4-dichloropyrimidines is regioselective and affords mainly the mono-phenyl product resulting from the reaction at the 4-position. We demonstrated that functional group manipulation directly on the organobismuth species is possible and gives access to functionalized aryl groups that are challenging to transfer such as formylaryl groups. The cross-coupling of a formylphenyl group was efficiently accomplished by using our protocol. Studies on the derivatization of organobismuthanes and their use in cross-coupling reactions with other electrophiles are in progress in our laboratory. Results will be reported in due course.

Experimental Section

Typical Procedure for the Cross-Coupling Reaction between Organo-bismuthanes and 2-Chloro-, 2-Bromo-, and 2-Triflyl(di)azines: In a sealed tube, the 2-halo(2-triflyl)azine or -diazine (0.30 mmol) was dissolved in dry *N,N*-dimethylformamide (4.5 mL). Cesium carbonate (0.60 mmol) was added, followed by tetrakis(triphenylphosphine)palladium (0.015 mmol) and the arylbismuth reagent (0.12 mmol). Argon was bubbled into the reaction mixture for 5 min. The tube was sealed and heated at 130 °C overnight. The reaction mixture was cooled to room temperature, diluted with a saturated aqueous solution of sodium hydrogen carbonate (50 mL), and extracted with ethyl acetate (2 × 50 mL). The combined organic phases were washed with a saturated aqueous solution of sodium hydrogen carbonate (2 × 50 mL) and brine (2 × 50 mL), dried with sodium sulfate, filtered, and concentrated under reduced pressure. The crude product was purified by flash column chromatography on silica gel (hexanes/ethyl acetate) to afford the corresponding product.

CCDC-949485 (for **20**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Supporting Information (see footnote on the first page of this article): Experimental details, characterization data, and copies of the ¹H and ¹³C NMR spectra of all key intermediates and final compounds.

Acknowledgments

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